



# Methodological Foundations of the Traffic Simulation Process

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**Abstract** There is an urgent need to design and evaluate strategies that improve the mobility of people and goods in urban regions while reducing energy consumption and vehicular emissions. Traffic simulation is an increasingly powerful tool to address this challenge; however, it requires extensive input data and involves multiple steps and decisions, which can make the process complex. This paper provides researchers and practitioners with a comprehensive methodological framework for modeling and simulating urban traffic, along with key insights and practical recommendations. These contributions are grounded in both the literature and the authors' experience across multiple traffic simulation applications, enabling the paper to offer guidelines that facilitate the implementation of urban traffic simulation while identifying areas that warrant further research. Unlike previous reviews, this study synthesizes practical lessons from real-world microsimulation projects and integrates methodological guidance applicable across different simulation scales.

DOI: <https://doi.org/10.34196/ijm.00342>

## 1. Introduction

Traffic simulation is increasingly used by companies and institutions to evaluate scenarios and analyze urban mobility systems. A wide range of software tools support these applications, including SUMO, PTV VISUM, and PTV VISSIM. From a modeling perspective, substantial progress has expanded the available approaches for simulation and analysis. Practical guidelines developed in the early 2000s, such as the toolkits by the US Department of Transportation (Alexiadis et al., 2004; Dowling et al., 2004; Holm et al., 2007), remain widely used; however, they do not fully integrate current technical developments and require updates to reflect advances achieved over the past two decades. Methodologically, numerous approaches capture specific aspects of reality, yet recent developments have blurred the distinctions between approaches, tools, and applications. Although several toolkits exist, few integrate scale selection, data availability, and energy indicators within a unified framework.

Therefore, this study examines traffic simulation by outlining key steps and guidelines to support its practical application, while providing a basis for further development throughout the simulation process. The document follows a structured, step-by-step approach. It begins with an overview of traffic simulation and common modeling approaches, and then reviews available software tools, including a comparative discussion of their characteristics, applications, computational requirements, and key features relevant to implementation.

The study aims to serve as both a methodological framework and a practical guide for urban traffic simulation. It presents a software-independent approach that supports the analysis of mobility patterns and the evaluation of strategies across multiple spatial scales. In addition, by incorporating

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**Author Keywords:** smart mobility, activity-based modeling, transportation planning, urban traffic modeling, microsimulation

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**Table 1.** Classification of traffic simulation models by scale

| Scale       | Level of detail                                                          | Region size                                                    | Mobility modeling                                                                      | Assumptions                                                     | Modeling principles                                                           | Underlying algorithms                                 | Software options                                             |
|-------------|--------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------|
| Microscopic | Each vehicle                                                             | Small city districts (e.g., freeway, arterial, corridor, etc.) | Individual dynamics of vehicles in intersections, specific roads, or small city zones. | Individual vehicles make decisions to get to their destination. | Individual mobility decisions in networks will represent the mobility system. | Longitudinal model, Lateral model, intersection model | Aimsun, Transmodeler, SUMO, PTV VISSIM, MATSIM, Cube Dynasim |
| Mesoscopic  | Each vehicle. Movement is ruled by the average speed on the travel link. | City districts                                                 | Flows of vehicles between city zones by type of activity.                              | Vehicles moving in queues                                       | Queues of individual vehicles that follow shockwave dynamics.                 | Longitudinal model, intersection model                | Aimsun, Transmodeler, Mezzo Polaris, SUMO Meso               |
| Macroscopic | Average vehicle dynamics, such as traffic density and average speed.     | Entire cities                                                  | Flows of traffic between largely aggregated zones.                                     | Aggregated vehicles flowing by main roads.                      | Aggregated vehicles flowing through roads based on fluid particle dynamics.   | Flow model                                            | Aimsun, Transmodeler, PTV VISSUM                             |

Alexiadis et al., 2004; Algherbal and Ratrou, 2025; Azam et al., 2019; Lopez et al., 2018.

vehicle energy efficiency-based metrics within a broader mobility perspective, it shifts the focus from vehicle-centered analyses toward the characterization of people's mobility patterns.

The manuscript is organized as follows. It first addresses simulation scale, defining scope and granularity, followed by a modeling approach for estimating travel demand using structured statistical procedures. It then analyzes simulation software tools, highlighting their strengths, limitations, and appropriate contexts. Recent advances are incorporated throughout, and the study concludes by identifying areas for future research.

## 2. Traffic Simulation Framework

Recognizing that data type, data quality, and resource availability critically influence the selection of an appropriate simulation model, this section first reviews the classification of traffic simulation models by scale, then introduces a general methodological framework applicable across scales, and finally outlines the fundamental principles for representing driving patterns.

### 2.1. Traffic Simulation Scales

Simulation models are classified into three categories, microscopic, mesoscopic, and macroscopic, according to their spatial and temporal resolution, as summarized in **Table 1**.

The selection of an appropriate modeling scale for a given application depends on multiple factors, including the required level of detail, availability and granularity of input data, the traffic performance indicators to be evaluated, computational resources and time constraints, capabilities of the simulation platform, and the scope of the decision-making objectives. For example, (Varga et al., 2023) highlights that the mesoscopic scale often balances resolution and computational cost, making it suitable for corridor-level applications instead of full network microsimulation.

*Microscopic simulation models*, or just microsimulation, represent traffic at the level of individual vehicles, capturing movement through dynamic variables such as position and velocity and explicitly modeling driver behavior and interactions (Alexiadis et al., 2004; Janson Olstam and Tapani, 2004; Nor Azlan and Md Rohani, 2018). Powered by car-following and lane-changing models, their main strength lies in the detailed representation of the traffic, enabling a multidimensional analysis. Examples are: localized infrastructure design, such as the evaluation of multiple design scenarios of a multi-lane roundabout in a small-scale context (Guerrieri and Khanmohamadi, 2024); the analysis of congestion dynamics (Uthpala et al., 2023), traffic management strategies, such as the effect of path widths by modeling cyclists' complex behaviors like maneuver decisions and lateral movements, into the bicycle flow capacity (Brunner et al., 2024) or the analysis of bicycle flow characteristics

in relation to infrastructure design (*Pérez Castro et al., 2025*); the impact of driving risk when a shockwave propagates in a queue at signalized intersections (*Wang et al., 2025*); the impact of an imperfect perception on the traffic flow dynamics, to be used in the study of mixed traffic of human and automated driving (*Postigo et al., 2025*); and the implementation of Intelligent Transportation Systems (ITS) (*Alexiadis et al., 2004*). This level of detail, however, entails high computational and data requirements, which limit the spatial extent of applications. A recent review of advancements in traffic simulation for road safety highlights the need for refined models that capture real-world traffic dynamics to improve predictive capabilities, further enhanced by integrating other models and machine learning algorithms (*Mustapha et al., 2024*).

At the other end of the spectrum, macroscopic models describe traffic using aggregate variables such as vehicular density, average flow, and the mean speed of a traffic stream (*Alexiadis et al., 2004; Ding, 2011*). These models operate at the level of road segments rather than individual vehicles and do not explicitly represent behavioral components such as trip generation or mode choice, and validation involves replication of observed congestion patterns (*Krista et al., 2004*). Their reduced complexity allows the analysis of large-scale systems, including entire urban areas, and facilitates the use of aggregated data sources for calibration and validation. These models offer great capability to understand and make decisions on a large scale for strategic planning and the evaluation of network-wide conditions.

Between these two levels, mesoscopic models provide an intermediate representation by simulating individual vehicles (i.e., emissions, travel times, etc.) while governing their dynamics through aggregate relationships (i.e., speed-density or flow-density relationships that can be defined with few parameters) (*Aimsun, 2025; Holm et al., 2007*). This balance between detail and computational efficiency makes them suitable for applications at the corridor or subnetwork level. As examples, a scenario simulation framework based on vehicle trajectory big data was used to evaluate the impact of shifts in residential travel modes on carbon emissions, identifying peripheral areas as having the most significant carbon reduction potential (*Li et al., 2025*); and a mesoscopic traffic simulation model was developed to assess the vulnerability of road infrastructure in an urban area to closure scenarios caused by flooding (*Marian et al., 2024*).

## 2.2. Traffic Behavior Models

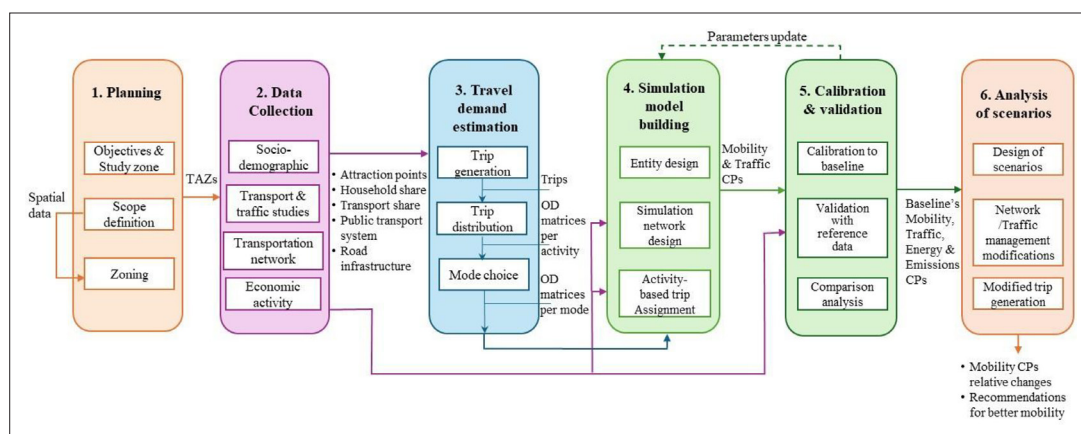
Key traffic behavior models, mainly considered in microscopic and mesoscopic simulations include gap-acceptance, speed adaptation, lane-changing, ramp merging, overtaking, and car-following (*Janson Olstam and Tapani, 2004*). Gap acceptance models determine the minimum acceptable spacing between vehicles, while speed adaptation refers to the adjustment of a vehicle's speed at a given network location to conform to the road's design speed. Lane-changing models describe drivers' behavior governing lateral movements between lanes, and finally, car-following models describe drivers' behavior governing longitudinal movements along the road by linking a vehicle's motion to its interaction with preceding vehicles. Movement is determined by factors such as desired, average, and maximum speeds, responses to leading vehicles, and the conditions under which lane changes occur when necessary or feasible.

In mesoscopic simulations, car-following models typically neglect acceleration and deceleration, assuming that vehicles are either stopped or traveling at their desired speeds. This simplification can be significant in applications where traffic control devices strongly influence flow. In such cases, the absence of acceleration dynamics may lead to flow overestimation, which must be compensated through calibration of other parameters. As a result, mesoscopic simulation is primarily applied to large-scale road networks, offering reduced effort in network construction and calibration compared to microscopic models, particularly when aggregate flow effects can be adequately represented through parameter adjustment (*Aimsun, 2025; Sun et al., 2020*). *Sun et al. (2020)* summarized recent applications of mesoscopic simulation for assessing the impact of transportation strategies and system response to emergency scenarios in relatively large urban zones.

**Table 2** presents a summary of the fundamentals of the most common car-following models used in traffic simulation tools. They are classified in models based on the safe distance premise to avoid collisions, models that consider the drivers' perception and their reactions to stimuli from the vehicle ahead, and other more recent approaches. For reference, SUMO includes Krauss model as the default

**Table 2.** Car-Following traffic models

| Class                                | Models                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gap acceptance                       | Gipps model (Gipps, 1981)                                            | Based on the premise that each driver sets limits on their desired braking and acceleration rates. The speed is based on a safe following distance to avoid a possible collision with the vehicle ahead. The behavior is controlled with response times, break rates, and maximum desired speed. This model is widely preferred for simulation purposes (Ranjitkar and Kawamura, 2005).                                                                                                                   |
|                                      | Krauss model (Krauß, 1998)                                           | Alike but simpler than Gipps' model. A stochastic model based on the same assumption that vehicles move without colliding. The model is set up with the same parameters as in the Gipps model. It is implemented in the SUMO simulator (Kanagaraj et al., 2013; Ranjitkar and Kawamura, 2005).                                                                                                                                                                                                            |
|                                      | Gazis-Hermann-Rothery (GHR) model                                    | Referred to as the general car-following behavior. The relationship between a leader and a follower vehicle is a stimulus-response type of function where the follower vehicle's acceleration is proportional to its speed, the speed difference with respect to the leader, and the space headway. This type of model assumes that the follower reacts to arbitrarily small changes in the relative speed (Janson Olstam and Tapani, 2004).                                                              |
| Psycho-physiological driver behavior | Leutzbach and Wiedemann model (1986)                                 | It takes into consideration the influence of the driver's perception on velocity control. A driver can be in different driving modes, such as free driving, approaching, following, and braking. The driver switches from one mode to another as soon as they reach a certain threshold, which can be expressed as a function of speed difference, space headway, etc. Versions of this model are used in VISSIM. (Kanagaraj et al., 2013; Ranjitkar and Kawamura, 2005).                                 |
|                                      | Fritzsche model (1994)                                               | This model also considers human perception in defining the model regimes. For example, there is a threshold for the minimum speed difference that the driver perceives. In addition, it incorporates four thresholds for the follower's space headway to its leader: Desired distance, the risky distance, the safe distance, and the braking distance (Janson Olstam and Tapani, 2004).                                                                                                                  |
| Cell based model                     | Nagel-Schreckenberg model (cellular – automata) (Nagel et al., 1998) | A computationally efficient stochastic model that simulates the traffic flow using only integer operations. It incorporates imperfections of driving using a noise term in the update rules. The model deals with single-lane traffic flow of cars moving in a one-dimensional cellular chain under periodic boundary conditions, which considers the desired maximum speed, vehicle acceleration, deceleration, random delays, and update of vehicle location (Liu, 2012; Ranjitkar and Kawamura, 2005). |
| Trajectory based model               | Newell model (2002)                                                  | Based on the premise that the driver of the following vehicle drives as a shifted space trajectory of the leader vehicle. The space trajectory of the following vehicle is the same as that of the leader vehicle except for a translation in space and in time. It is driven by a time lag and a distance lag (Ranjitkar and Kawamura, 2005).                                                                                                                                                            |
|                                      | Intelligent Driver Model (IDM)                                       | IDM is a car-following model that describes the dynamics of the positions and velocities of single vehicles using ordinary differential equations (Gora et al., 2020).                                                                                                                                                                                                                                                                                                                                    |



**Figure 1** . Proposed methodology for implementing traffic simulation at any scale, aiming at evaluating strategies to improve mobility in a given region.

option together with alternatives such as the Intelligent Driver Model (IDM) and Wiedemann-like variants, while VISSIM relies on Wiedemann models.

### 3. Methodological Framework for Traffic Simulation

Based on the literature review and the authors' experience in applying traffic simulation to evaluate mobility strategies, a comprehensive framework is proposed. **Figure 1** outlines the process and illustrates its data-intensive nature, highlighting its strong dependence on the amount, quality, and type of data.

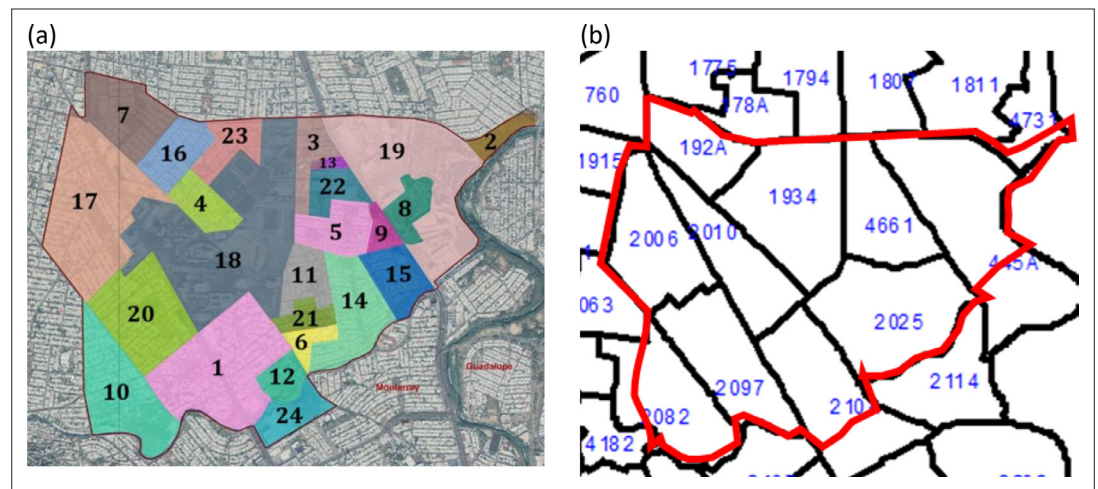
The framework consists of sequential steps. The first step, planning, defines the study's objectives and scope, including the region of interest in terms of location, homogeneous sociodemographic zones, and available physical infrastructure such as the road network, public transport, vehicle fleet, and energy sources for passenger and freight transport. The second step, transportation data, involves the development, acquisition, or post-processing of the origin–destination matrix, supported by the identification of travel needs and major trip attraction centers. The third step, travel demand estimation, specifies the set of trips within the study region, including origin, destination, departure time, route, and mode of transport. The fourth step, simulation, represents the dynamic execution of these trips over time. The fifth step, calibration and validation, assesses the model's ability to reproduce observed traffic conditions, with calibration involving parameter adjustment to improve accuracy (*Dowling et al., 2004*). Finally, the model is applied according to the objectives defined in the planning stage.

### 3.1. Planning

### 3.1.1. Scope Definition

The planning of a traffic modeling and simulation project is guided by the identification of key elements: the area of interest defining the network to be analyzed; the intended use of the model to evaluate traffic behavior, assess new strategies, or monitor performance; the required accuracy, defined as the degree to which model outputs match observed or accepted values; the key performance indicators used to support decision making; the required precision, understood as the unit of analysis for both inputs and outputs, such as individual vehicles, road segments, or zones; and the level of detail or resolution of the model; and the available resources, including expertise, software, time, computational capacity, budget, and data quality (*Krista et al., 2004*).

Regarding intended use, and following **Krista et al. (2004)**, traffic simulation models support improved decision making in transportation management, the evaluation and prioritization of planning and operational alternatives, reductions in design time and costs, minimization of traffic disruptions, communication with stakeholders, operation of existing roadway capacity, performance monitoring, and recently, as an intermediate layer of cities' digital twins (**Huertas et al., 2026**).



**Figure 2** . Zoning examples of the same area: District Tec in Monterrey, Mexico.

(a) TAZ as neighborhoods, (b) TAZ as AGEs. Note: (a) Original map is a public consultation document from the Partial Urban Development Program of Distrito Tec (*ImplancMTY*, 2021) (b) Original map is a public consultation document, retrieved from <https://www.coneval.org.mx/>.

### 3.1.2. Zoning

Once the area of interest has been defined, the next step is to divide it into zones (e.g., households and businesses) with uniform characteristics. This process, known as zoning (*Ortúzar and Willumsen, 2011*) produces Transportation Analysis Zones (TAZs). Zoning typically involves analyzing activity centers, sociodemographic characteristics, and the transportation system to define appropriate zone boundaries.

In transport modeling, considerable effort is often devoted to subsequent stages, while zoning is treated largely as a matter of “common sense.” However, arbitrary TAZ boundaries can introduce statistical and geographical errors, becoming a major source of inaccuracy in the modeling process (*Martínez et al., 2009*). TAZ definitions shape the socioeconomic characteristics of subareas and, consequently, influence trip OD patterns. When TAZs lack homogeneity and compactness, trip generation and OD matrices may be poorly estimated. Moreover, predefined zoning systems may fail to capture the dynamic spatial and temporal evolution of land use.

**Figure 2** shows an example of different zoning approaches for District Tec in Monterrey, Mexico. Figure 2a delimits District Tec and defines TAZs as “neighborhoods”, with political and administrative distinctions, while Figure 2b delimits District Tec in red line over the Basic geostatistical areas (AGEs) defined by the National Institute of Statistics and Geography (INEGI) in Mexico. Zoning by neighborhoods can effectively represent mobility patterns, whereas zoning based on AGEs, with extensive statistical information available at this aggregation level, facilitates integration with broader urban simulation layers, such as digital twins, as in *Huertas et al. (2026)*.

Zoning approaches that determine the number and size of TAZs have generally evolved into four categories. The first is constraint-based zoning, in which each TAZ must satisfy modeling requirements such as compactness, population thresholds, or computational efficiency. Early spatial aggregation methods followed this approach by hierarchically grouping areas according to the similarity of selected attributes. The second is model-based zoning, which uses computational and transport-demand models to reduce forecasting inaccuracies and improve model performance. The third is criteria-based zoning, where zones are delineated according to planning, land-use, socioeconomic, and network-related characteristics. More recently, data-driven zoning has emerged, applying clustering techniques, machine learning, and large-scale mobility datasets to generate spatially homogeneous zones based on observed travel behavior and urban dynamics. **Table 3** summarizes the main characteristics of some classical, widely recognized, and also recent zoning studies in each category.

**Table 3.** Zoning methods

| Type             | Author                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Constraint based | (Openshaw, 1977)              | Maximization of the statistical precision of the estimation of the OD matrix cells.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                  | (Baass, 1981)                 | Trip generation/attraction homogeneity<br>Adjustment of TAZ boundaries to political, administrative, or statistical ones.<br>Minimization of intra-zonal trips                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                  | (O'Neill, 1991)               | Trip generation/attraction homogeneity<br>Contiguity and convexity zones<br>Compactness of TAZ shapes<br>Exclusiveness (no doughnuts or islands) of zones<br>Equity in terms of trip generation (small standard deviation across zones).<br>Adjustment of TAZ boundaries to political, administrative, or statistical ones.<br>Respect for physical separators<br>Decision-makers' preferences are considered when defining the number of TAZs.                                                                                                                                                                                                                                                                        |
|                  | (Prosperi et al., 2021)       | Traffic Zones Discretization and Origin-Destination Matrix Estimation by means of Fusion.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Model based      | (Luo and Zhang, 2021)         | TAZ are defined by combining static zoning and dynamic zoning, by adjusting the zones based on real-time traffic, enabling a maximum flow control strategy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Data-Driven      | (Martínez et al., 2009)       | TAZs are defined based on actual travel demand patterns, using a density-smoothing algorithm.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                  | (Yang et al., 2022)           | Clustering and GIS-based spatial analysis is used for grouping spatial units with similar mobility characteristics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Criteria Based   | (Crevo, 1991)                 | Minimize prediction errors associated with the spatial aggregation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                  | (Ortúzar and Willumsen, 2011) | The aggregation error caused by the assumption that all activities are concentrated at the centroid of the TAZ should not be too large. Each zone must be compatible with other administrative divisions, particularly with census zones. TAZ should be as homogeneous as possible in terms of land use or composition. TAZ boundaries must be compatible with cordons, screen lines, and those of previous zoning systems. Avoid using main roads as zone boundaries.<br>The zone configuration should facilitate the clear definition of centroid connectors. Zones do not have to be of equal size; if anything, they could be of similar dimensions in travel time units (i.e., smaller zones in congested areas). |

## 3.2. Transportation Data Collection

Once TAZs have been established, the next stage in the traffic simulation modeling process is data collection and preparation. These data support subsequent steps, including transport demand modeling, simulation model development, and model calibration and validation. The process involves gathering available information and, when necessary, generating data through field campaigns, acquiring it from open or commercial databases (*Chaparro Sierra and Huertas, 2024*) or estimating missing variables such as attraction-zone capacities. The data are then processed for model implementation. One of the main limitations in traffic simulation arises at this stage when data quality is insufficient and resources for additional collection are limited. In such cases, the use of less data-intensive tools or broader modeling scopes was recommended (*Alexiadis et al., 2004*). Nowadays, machine learning techniques can estimate mobility patterns even from incomplete data. For example, INFOSTOP, a two-step stay-point detection algorithm for mobility trajectory analysis, was applied in the OD matrix construction presented by *Chaparro Sierra and Huertas (2024)*.

**Table 4** summarizes the required data, whose availability should be verified, as each component directly influences simulation accuracy and outputs. In general, traffic simulation data can be grouped into three categories: input data for the transport demand model, parameters for simulation model development, and driver-behavior characteristics used during simulation.

When developing a transport demand model, the required information depends on data availability and sources. In the best-case scenario, OD matrices are already available, substantially reducing data collection efforts. The main input components are: (a) modal preferences and travel habits, describing travel behavior between zones; and (b) the location, capacity, and activity centers that support the identification of trip generation and attraction areas within the defined TAZs. These inputs can also support the refinement of TAZ boundaries to better represent observed mobility patterns.

The second classification includes the information required for the simulation model to operate properly. It mainly comprises two components: (a) traffic network and transport systems, which support trip assignment and journey's trajectories definition within the transport model; and (b) calibration and validation data, typically based on key performance measures such as traffic volume, level of service, travel and delay times, capacity, and queue lengths.

A third category of data, increasingly available through current technologies, consists of massive and passively collected mobility information from GPS-enabled mobile devices (*Chaparro Sierra and Huertas, 2024; Eom et al., 2026*). These data provide representative samples of individual mobility patterns with high spatial and temporal resolution, focusing on people rather than vehicles. Combined with emerging machine learning techniques for large-scale data processing, they offer a promising alternative to traditional approaches, addressing limitations such as high costs, limited representativeness, and low spatial and temporal resolution.

The fourth data type relates to driving characteristics (DCs). Depending on the simulation software, additional parameters may be required to better represent real-world driving conditions, as variations in driving behavior can significantly affect simulation outputs. For instance, traffic characteristics may differ across TAZs. Indicators such as modal split and accident reports can provide insights into driver aggressiveness, the use of dedicated or high-occupancy vehicle lanes, and trip-generating activities, supporting more accurate parameter tuning.

## 3.3. Travel Demand Estimation

Travel demand models are widely used to predict travel demand and analyze mode, activity, and destination choices based on current conditions. However, they are not intended to evaluate the dynamic interactions between vehicles and transportation systems. Within travel demand estimation, the literature mainly distinguishes between two research directions: trip-based and agent-based (or activity-based) modeling (*Harb et al., 2022; Nguyen et al., 2025*).

### 3.3.1. Trip-based Travel Estimation

Trip-based modeling focuses on trip generation as the primary unit of analysis, representing travel demand through aggregated regional mobility patterns. As the classical approach, it models individual daily travel as independent origin–destination trips derived from statistical and census-based data, typically provided by from government institutions, public agencies, or organizations managing

**Table 4.** Required input data

| Type     | Example                        | Description                                                                                                                                                                                                                                                                                    | Used in: |    |           |
|----------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----|-----------|
|          |                                |                                                                                                                                                                                                                                                                                                | TM       | SM | DC        |
| Network  | Road geometry and capacity     | Number of lanes, designated turning lanes, length of lanes, free flow speed.                                                                                                                                                                                                                   |          | X  |           |
|          | Traffic controls               | Traffic signals' location, signal timing, etc. Also known as geometric data, it consists of the number of lanes, designated turning lanes, length, and free flow speed. Also, it may include data on traffic control devices, such as the location of traffic signals and their signal timing. |          | X  |           |
|          | Level of service (LoS)         | The measure of the quality of flow through a roadway segment or an intersection.                                                                                                                                                                                                               |          | X  |           |
|          | Location of activity enters    | Predefined TAZ (attraction and generation zones).                                                                                                                                                                                                                                              | X        |    |           |
|          | Transit system                 | Public transport information (type, lines, stops, users, etc.)                                                                                                                                                                                                                                 | X        | X  |           |
| Vehicles | Vehicle mix                    | Composition of the vehicle fleet in the study area.                                                                                                                                                                                                                                            | X        | X  |           |
|          | Maximum acceleration rate      | The performance obtained from vehicle characteristics like weight and power varies according to the type of vehicle. In general, for lower speeds, the acceleration rates are greater.                                                                                                         |          | X  |           |
|          | LoS of vehicles                | Capacity (number of passengers), idle capacity.                                                                                                                                                                                                                                                |          |    |           |
|          | Vehicle lengths                | It is the average length of the vehicles in the study area, which may vary according to the region and the needs of the population.                                                                                                                                                            |          | X  | X         |
|          | Volume studies                 | The annual average daily traffic: Count location and duration using physical counting. It can be done with inductive loop, magnetic, pneumatic road tubes, active or passive infrared, microwave radar, ultrasonic, passive acoustic or video image processing.                                |          | X  |           |
| Traffic  | Directional volume studies     | Entry volumes, turning volumes, and turning movements at intersections.                                                                                                                                                                                                                        |          | X  | X         |
|          | Demand                         | Traffic reports (modal preferences, travel habits, OD matrices, accident reports)                                                                                                                                                                                                              | X        | X  | X         |
|          | Travel time and delays studies | Average time that it takes to go from point A to B through studies requiring a test vehicle (floating-car technique, average speed-technique, moving vehicle techniques) or not (license-plate method and the interview method).                                                               |          | X  |           |
|          | Capacity flow data             | Maximum volume per segment, maximum entrance rate, maximum (saturation) flow.                                                                                                                                                                                                                  | X        | X  |           |
|          | Temporal location              | GPS position at a given time                                                                                                                                                                                                                                                                   | X        |    |           |
| People   | Socio-demographics             | Age, education, economic activity, income, family composition, etc.                                                                                                                                                                                                                            | X        |    |           |
|          |                                |                                                                                                                                                                                                                                                                                                |          |    | Continued |

**Table 4.** Continued

| Type                    | Example            | Description                                                        | Used in: |    |    |
|-------------------------|--------------------|--------------------------------------------------------------------|----------|----|----|
|                         |                    |                                                                    | TM       | SM | DC |
| Driving characteristics | Aggressivity level | Average speed, free speed, and other safety perception parameters. |          | X  | X  |
|                         | Modal split        | Proportion of mode and type of cars used by drivers.               | X        | X  | X  |
|                         | Average occupancy  | Use of HOVs, trip generating activities.                           |          | X  | X  |
|                         | Accident rates     | Frequency, location, time of the day of incidents and accidents.   |          | X  | X  |

TM = Transport demand modeling, SMB = Simulation model building, DC = driving characteristics.

publicly available information. These datasets support the analysis of population distribution and demographic characteristics, making the approach suitable for evaluating long-term policies and broad mobility trends.

### 3.3.2. Agent (or Activity)-based Travel Estimation

Activity-based (or agent-based) modeling extends the trip-based approach by representing individuals as active agents whose travel behavior is shaped by attributes such as family status, occupation, and activity patterns. In this framework, trip generation becomes a more complex process aimed at capturing daily trip chains and the activities driving individual movements. By tracking daily trajectories within a broader system, it provides greater granularity and more dynamic insight into short-term behavioral changes than aggregated population-based models. **Table 5** summarizes both approaches.

At a more detailed level, activity-based modeling represents trips as OD journeys, often structured as simple chains from home to a primary destination (e.g., work or commerce) and back home. However, real behavior frequently departs from this structure. After reaching a primary destination, individuals may include secondary stops (e.g., errands or drop-offs) or combine multiple intermediate purposes in stochastic ways. This expanded representation enables more realistic modeling of trip chains with variable intermediate stops and path dependencies (*Krista et al., 2004*).

Examples of activity-based models include the evaluation of multimodal transportation shares for reducing traffic congestion (*Uthpala et al., 2023*), the development of an open-source agent-based traffic simulation tool using cellular automata (*Górka and Malecki, 2024*), the analysis of sudden travel behavior changes in mixed traffic flow with automated vehicles (*Wang et al., 2025*), and large-scale pedestrian movement modeling (*Kaziyeva et al., 2023*).

This richer trip definition has several implications: it enables simulation models to incorporate coupling between stop scheduling and mode choice, time flexibility, and route reshaping in response to intermediate opportunities. In doing so, it also better reflects observed travel diaries, allowing microsimulation engines to allocate more realistic time budgets across activities. Integrated frameworks such as the BayABM model combine Bayesian networks with activity-based architectures to link choice and chaining behavior at the individual level (*Luo et al., 2024*). Likewise, project-based

**Table 5.** Transport demand models

|                        | Trip-based approach (TBA)                                                | Activity-based approach (ABA)                                                            |
|------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Scale                  | City or large urban region                                               | Urban district or smaller                                                                |
| Type of simulation     | Macroscopic                                                              | Mesoscopic or microscopic                                                                |
| Assumption about trips | Production and attraction of trips by zone (Ortúzar and Willumsen, 2011) | Trips generated and constrained by the activities of individuals                         |
| Spatial assumptions    | Each zone only contains the centroid.                                    | Zones contain individual households and activity centers (Codeca, 2022)                  |
| Approach               | Individual, not related trips (Ortúzar and Willumsen, 2011)              | Interrelated tours and trip chains                                                       |
| Required data          | Zone aggregated data                                                     | Individual data by household and individual attributes                                   |
| Comments               | Data must be complete for each zone; fewer approximations are acceptable | Approximations at the individual level can lead to a good representation by aggregation. |

activity-agent microsimulation models have recently been proposed to overcome the limitations of simple trip chaining, enabling flexible representation of multiple stop patterns across days (Miller, 2024).

Recent research in large-scale activity-based demand generation (Nguyen et al., 2025) refines the generation of chained activity tours beyond single-trip cycles. Similarly, Moeckel et al. (2024)'s ABIT framework (2024) extends weekly activity patterns to capture habitual sequences beyond simple home-work-home loops. Moreover, Harb et al. (2022) integrate multi-purpose chaining in short-term travel demand modeling to better represent mixed trip behaviors, such as combining business and shopping stops within a single tour. Figure 3 exemplifies representations of single chains (TBA approach) versus more complex activity chains (ABA approach).

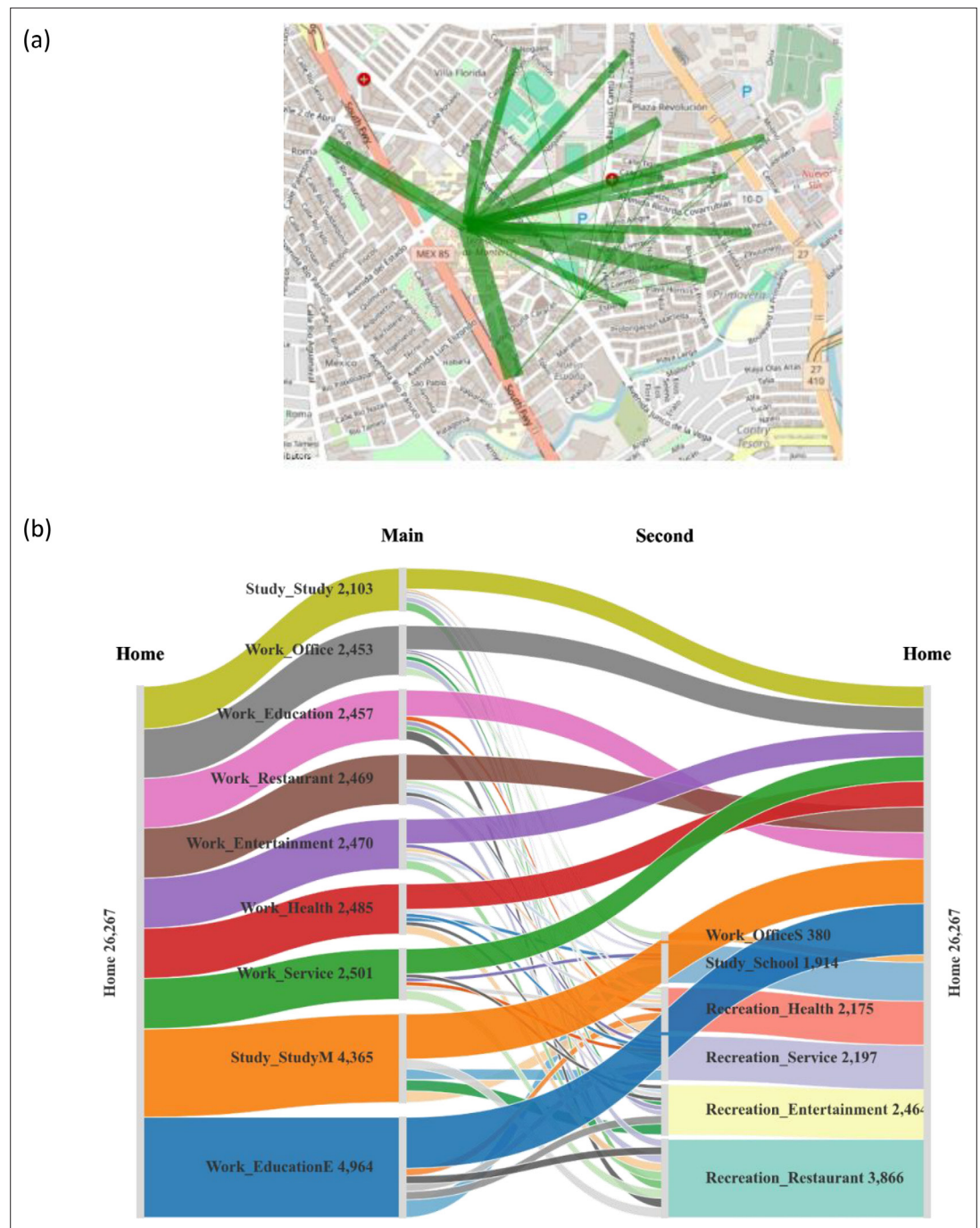
The primary input for these models is population survey data, based on the assumption that most regional mobility originates from the individuals who inhabit the area. Using statistically representative samples, travel habit surveys collect information that can be mathematically translated into regional trip patterns (Tamin and Willumsen, 1989). Consistent with the two modeling approaches, surveys may also be trip- or activity-based. Trip-based surveys focus on trips generated by households and aggregate them into travel volumes, as in Díaz-Ramírez et al., 2023, whereas activity-based surveys emphasize the activities that motivate movements across locations and times (McNally, 2008) as in Eom et al. (2026).

Survey data are then processed to estimate origin-destination (OD) matrices, one of the most critical inputs for simulation models. These matrices may vary in aggregation level and can be static or time-dependent (e.g., by day or time period). As shown in Table 6, OD matrix estimation remains essential, even when not explicitly required by simulation software.

### 3.3.3. The Four-stage Travel Demand Model

Travel demand can be modeled through a four-stage framework Ortúzar and Willumsen (2011) for trip-based modeling, whose stages are described below.

*Generation of trips stage:* This stage of the transport model estimates the total volume of trips generated and attracted by each TAZ in the region. Depending on modeling objectives, trips are typically classified by purpose and origin (home-based or non-home-based) (Krista et al., 2004). Two complementary models are developed: one for trip production and another for trip attraction. In the production model, trips are assumed to originate from residential zones, operating at the household level. Trips to work and school destinations are represented at the zone level within the attraction model. A closed-system assumption is commonly adopted when external mobility is considered negligible. Additionally, many mobility studies incorporate an urban planning component, requiring long-term analysis to anticipate changes in regional travel demand. Since the activity-based approach relies on the disaggregated analysis of generation, TAZs are divided into households and workplaces. Its implementation requires explicit spatial considerations for assigning trip origins and destinations.



**Figure 3** . Examples of OD matrices representations (a) single trips, one mode, (b) activity chains.

*Distribution:* At this stage, the mobility relationships between zones are defined through spatial linkage. All trips generated in the previous stage are assigned to an origin and a destination zone, which may be the same zone, an adjacent zone, or a more distant zone (Krista et al., 2004). In general, distribution models assume that travel time is perceived as undesirable; consequently, trips tend to concentrate in nearby areas and progressively decrease with increasing distance. In the ABA, distributing travel demand requires accounting for disaggregate user characteristics, such as access to work or school opportunities that satisfy individual needs, including those within the same household. This stage also incorporates the spatial structure of the network and its alignment with user requirements.

**Table 6.** Different formulations of OD matrices required in simulation models

| Input for simulation software                                                          | Description                                                                                                                                 | Transformation to OD matrix                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| # Trips from centroid to centroid once a day                                           | Software packages may require input trips that go from one location to another.                                                             | This is a direct formulation of the OD matrix, which could result from the generation stage of the transportation model, and can be requested as an input for each location.                                                                                    |
| # Trips from centroid to centroid over periods of time (hourly, 15 minutes)            | Simulation software may require assigning trips at different time periods as the simulation resolution increases.                           | This is a formulation that is extended into a 3 <sup>rd</sup> dimension of time intervals (Fellendorf and Vortisch, 2010).                                                                                                                                      |
| # Trips from centroid to centroid of different modes over periods of time, etc.        | As more detail is used in the simulation, additional disaggregated information could be included as input for simulation software packages. | This formulation extends the previous OD matrices and may include several factors that differentiate the traffic as intended.                                                                                                                                   |
| Probability that a user has a specific location, activity chain, and mode of transport | Meso- and microscopic simulation software will require a broad set of highly disaggregated data.                                            | This formulation of the OD is disaggregated and may be obtained from the previously explained multidimensional OD matrix. It may require calculating the conditional probabilities of the individual's travel characteristics (Codeca, 2022; Eom et al., 2026). |

**Modal Choice:** This stage models the transport mode choice process used to execute the distributed trips from the previous stage. It is based on the concept of attractiveness or utility of alternatives within a user's choice set. Accordingly, modes with attributes better suited to specific user conditions have a higher probability of being selected to complete trips. These attributes may be evaluated at the individual or group level, based on shared location or characteristics, to estimate each mode's utility. Examples include travel time, travel cost, waiting time, and accessibility (e.g., proximity to stops or vehicle ownership). In the activity-based approach (ABA), modal choice is implemented through models that estimate the probability of selecting each mode using individual characteristics, such as access to transport options, household and workplace locations, and network spatial structure. This results in a disaggregate model that predicts mode choice at the individual level. Repeating this process across all simulated individuals enables the derivation of aggregate mode shares.

**Trip assignment:** The final step in travel demand modeling determines the trajectory users select to reach their destinations using the chosen mode. This requires explicit consideration of the network available for each mode. For urban road systems, congestion effects, roadway type, and connectivity to TAZs must be accounted for. Trip assignment within the ABA approach involves evaluating all feasible paths across the network, incorporating traffic conditions and travel times along the selected itinerary. An additional step is required to complete demand modeling: iterative feedback to the distribution and mode choice stages to explore alternative solution states and ensure consistency with the initial OD matrix. This process is referred to as equilibrium between the transportation system and its users.

Each stage may be modeled using different methods in a hybrid framework, adapting travel demand modeling to available data and expertise. **Codeca et al., 2015** developed a hybrid activity-based simulation model of Luxembourg using SUMO, combining count-based trip generation with agent-based distribution and mode choice, and applying maximum likelihood estimation for route assignment that minimizes individual travel times.

**Table 7** explains the methods reviewed in the literature related to travel demand modeling. Five categories are included: simple methods that update old OD matrices with new available information, theoretical methods that are based on laws of physics and phenomena from different disciplines, and are applied by analogy; and counting-based methods that are based on observations of flows and are based on assignment equations (**Bierlaire, 1997**); estimation methods that calculate unknown parameters of equations and functions that model demand (**Ben-Akiva and Bierlaire, 1999**); and machine learning methods that are algorithms used for multiple types of tasks that can model travel demand (**Salas et al., 2022**), which can be identified as the current trend.

**Table 7.** Methods for travel demand modelling

| Category               | Method                         | Approach |     | Model Stage |   |   |   | Level |    |    |  | Details                                                                                                                        | Sources                         |
|------------------------|--------------------------------|----------|-----|-------------|---|---|---|-------|----|----|--|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
|                        |                                | TBA      | ABA | G           | D | M | A | Mi    | Me | Ma |  |                                                                                                                                |                                 |
| Simple methods         | Growth factor                  | X        |     | X           |   |   |   |       |    | X  |  | Updating an old matrix using the actual number of trips.                                                                       | (Furnes, 1965)                  |
|                        | Tri- proportional              | X        |     | X           |   |   |   |       |    | X  |  | Growth factor considering cost distribution.                                                                                   | (Bierlaire, 1997)               |
| Theoretical methods    | Gravitational                  | X        |     |             | X |   |   | X     |    | X  |  | Analogy with Newton's gravitational law.                                                                                       | (Casey, 1955)                   |
|                        | Entropy maximization           | X        |     |             |   | X | X | X     | X  |    |  | Derived from the second law of thermodynamics.                                                                                 | (Ortúzar and Willumsen, 2011)   |
|                        | Agent-based                    |          | X   | X           | X | X | X | X     | X  | X  |  | Derived from cellular automata theory, using individual beha-vioral rules to define aggregate transportation.                  | (Zhang and Levinson, 2004)      |
| Counting-based methods | Proportional assignment        | X        |     | X           | X |   |   | X     | X  | X  |  | Consider that the flow proportion using a specific link is inde-pendent of the complete traffic flow.                          | (Dial, 1971)                    |
|                        | Restricted capacity assignment | X        |     |             | X |   |   | X     | X  |    |  | Based on Wardrop equilibrium (1952), taking congestion into account. The flow proportion depends on the complete traffic flow. | (Beckman et al., 1956)          |
|                        | Linear least squares           | X        |     | X           | X |   |   | X     | X  | X  |  | Fit the parameters of a model that minimizes the difference between estimated and real flows                                   | (Bierlaire, 1997)               |
| Estimation methods     | Maximum Likelihood estimation  |          | X   |             | X | X |   | X     | X  | X  |  | With a probability model and statistical hypotheses to make statistical inference.                                             | (Ben-Akiva and Bierlaire, 1999) |

Continued

**Table 7.** Continued

| Category                 | Method         | Approach |     |  | Model Stage |   |   |   | Level |   |   |   | Ma | Me | Details                                                                                                               | Sources               |
|--------------------------|----------------|----------|-----|--|-------------|---|---|---|-------|---|---|---|----|----|-----------------------------------------------------------------------------------------------------------------------|-----------------------|
|                          |                | TBA      | ABA |  | G           | D | M | A | Mi    |   |   |   |    |    |                                                                                                                       |                       |
| Machine Learning methods | Classification |          | X   |  |             |   | X |   | X     |   | X |   |    |    | Algorithms based on decision trees that optimize the modeling of mode choice based on decision rules.                 | (Hillel et al., 2021) |
|                          | Clustering     |          | X   |  | X           |   |   |   | X     |   | X |   |    |    | Unsupervised algorithms that group zones in the study region. Based on individual characteristics.                    | (Hafezi et al., 2019) |
|                          | Regression     | X        | X   |  | X           | X |   | X |       | X |   | X |    |    | Supervised algorithms that forecast the demand for travel, the distribution of activities of trips, and can forecast. | (Rocha et al., 2021)  |

TBA: Travel-based approach, ABA: Activity-based approach, G: Generation stage, D: distribution stage, M: Mode choice stage, A: Trip assignment stage, I: Microscopic simulation level, E: Mesoscopic simulation level, and A: Macroscopic simulation level.

**Table 7** illustrates how a single method can be applied across multiple modeling stages. This table also suggests the high potential for hybridization of the travel demand process by combining methods at each stage or using the same technique at different stages.

### 3.4. Building the Simulation Model

After defining the objectives and modeling approach, appropriate traffic simulation software must be selected. The platform should match the modeling approach, analysis scale, and project objectives, while model development relies on input data generated in earlier stages.

Open-source platforms include SUMO (Lopez et al., 2018), MATSim (Balmer et al., 2008), TRANSIMS, MOTUS, TRITONE, and OpenTrafficSim (Raju and Farah, 2021). Commercial alternatives include the PTV VISSIM and PTV VISUM platforms (Fellendorf and Vortisch, 2010), Aimsun (Casas et al., 2010), and CORSIM (Halati et al., 1997; Raju and Farah, 2021; Saidallah et al., 2016). Several studies have reviewed the evolution and capabilities of these simulation tools, while comprehensive catalogs and comparison resources are also available online (University of Leeds and ITS, 2000).

Raju and Farah (2021) reviewed the development of sixteen traffic microsimulation platforms throughout this century, identifying SUMO, VISSIM, and AIMSUN as the most widely adopted, followed by PARAMICS and CORSIM. SUMO and VISSIM have been extensively applied to evaluate the externalities associated with urban mobility policies and are increasingly used in emerging research on autonomous vehicles and connected mobility. More recent comparative studies have evaluated traffic simulation platforms according to modeling scale, licensing schemes, platform compatibility, visualization capabilities, application objectives, and support for intelligent transportation systems and connected or autonomous vehicles (Algherbal and Ratrout, 2025).

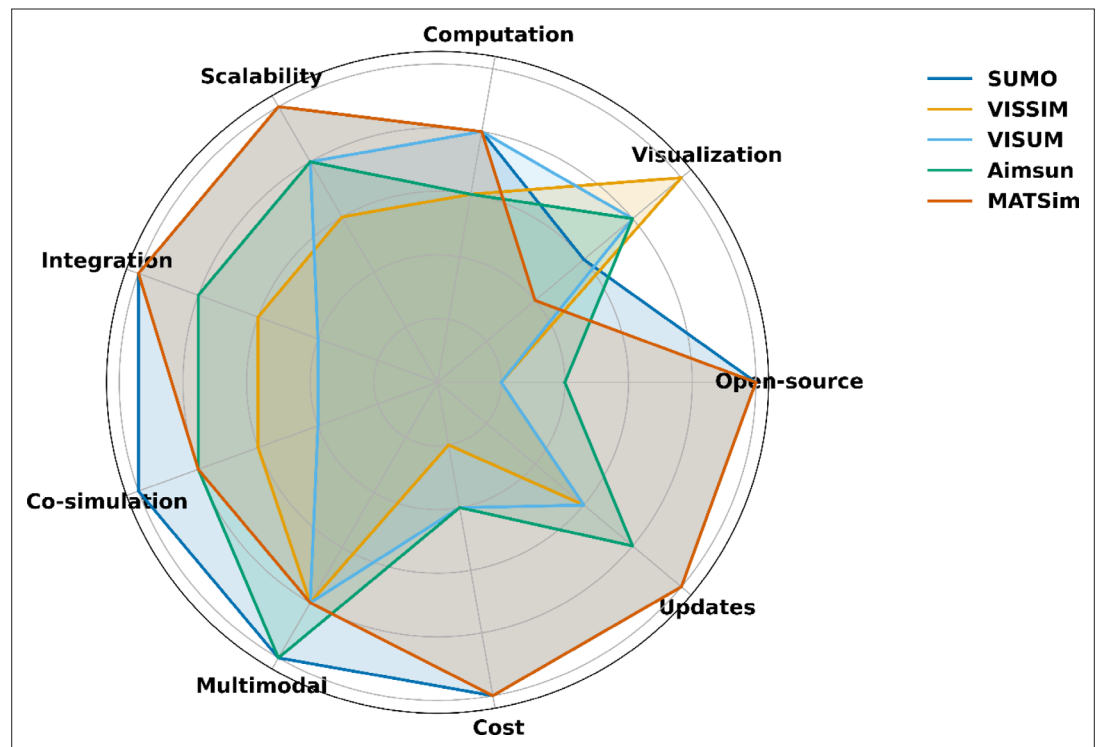
#### 3.4.1. Traffic Simulation Software

Although several reviews of simulation software are available, this study provides an independent comparison of existing tools to assess their suitability for traffic simulation applications. **Table 8** summarizes the main characteristics of these platforms.

Based on **Table 8**, widely used traffic simulation software exhibits distinct advantages depending on the application context. For projects with sufficient budgets and product-oriented requirements, such as government or privately funded transportation

**Table 8.** Most common simulation software tools

| Software     | Cost & License                                     | User-Friendliness                                 | Modeling Approach                         | Visualization                                            | Strengths (Power)                                             | Limitations                                 |
|--------------|----------------------------------------------------|---------------------------------------------------|-------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------|
| Aimsun       | Commercial, expensive; academic licenses available | User-friendly GUI, steep learning curve           | Microscopic, mesoscopic, hybrid           | Strong 2D/3D visualization                               | Flexible (multi-res), good for large urban projects           | Costly, proprietary                         |
| TransModeler | Commercial (Caliper Corp.)                         | GUI is okay but less intuitive than Aimsun/VISSIM | Micro + meso + dynamic traffic assignment | 3D visualization (moderate quality)                      | Strong for planning + operations integration                  | Less global user base, fewer academic users |
| SUMO         | Free, open-source                                  | Moderate (steep learning initially), script-based | Micro + meso                              | 2D GUI, limited native 3D (extendable via Unity/Blender) | Scalable (city-wide), great for AI, CAV, RL                   | Graphics not polished; scripting needed     |
| PTV VISSIM   | Commercial, very expensive                         | Very user-friendly GUI                            | Microscopic only                          | Excellent 2D/3D visuals                                  | Most realistic driver behavior (Wiedemann), industry standard | High cost, slower on very large networks    |
| PTV VISUM    | Commercial, costly                                 | GUI-focused, planner-friendly                     | Macroscopic (strategic planning)          | Visualization is more schematic than realistic           | Excellent for demand modeling, network assignment             | Not for micro-level traffic operations      |
| MATSim       | Free, open-source                                  | Code-based, not very beginner-friendly            | Agent-based, mesoscopic                   | Limited built-in; external tools for visuals             | Huge-scale simulations (whole regions)                        | Requires coding skills, less visual         |
| Cube         | Commercial (Bentley)                               | Planner-oriented GUI                              | Macroscopic (demand forecasting)          | Limited visualization                                    | Widely used for travel demand modeling                        | Not for microscopic operations              |
| Dynasim      | Commercial (PTV legacy)                            | Less common now                                   | Microscopic                               | Decent visualization                                     | Earlier detailed micro tool                                   | Mostly replaced by VISSIM                   |
| Mezzo        | Free (academic, KTH Sweden)                        | Basic, not very user-friendly                     | Mesoscopic (stochastic simulation)        | Very limited                                             | Efficient meso-sim, academic research                         | Not widely used, poor visuals               |
| Polaris      | Free (open source, Argonne National Lab)           | Requires coding                                   | Agent-based, mesoscopic                   | Limited                                                  | Strong research tool for large networks                       | Steep learning, low adoption                |



**Figure 4 .** Comparison of available traffic simulation software capabilities

infrastructure initiatives, commercial platforms including PTV VISUM and PTV VISSIM provide robust and comprehensive solutions. In contrast, SUMO is particularly well suited for research due to its open-source nature, active community, and collaborative ecosystem. From the authors' experience, SUMO provides high-quality outputs, supports multiple data formats for interoperability with external platforms, and offers Python-compatible modules that facilitate advanced research without licensing constraints, but requires extensive programming effort for data integration and customization. It can also be integrated with external applications such as Unity to improve visual representation.

**Figure 4** summarizes the key features identified in the analysis. Each axis represents a key evaluation dimension: Integration, the ability to connect with external data or tools; Visualization, the quality of graphical outputs; Scalability, the capacity to handle different network sizes; Cost, referring to licensing and expenses; Co-simulation, the ability to work with other simulators; Updating, the frequency of updates and level of support; Multimodal, the ability to model multiple transport modes; and Open-source, the availability of source code.

**Figure 4** compares some of the authors' highest-ranked and most widely adopted traffic simulation platforms, highlighting open-source tools such as MATSim and SUMO for their flexibility and continuous development, as well as commercial solutions such as VISSIM and VISUM, both developed by PTV Group, for their advanced visualization capabilities and extensive professional application (Algherbal and Ratrout, 2025).

The Appendix A summarizes recent applications of SUMO, MATSim, and VISUM reported in the literature.

### 3.5. Calibration and Validation

In traffic simulation, calibration consists of adjusting model parameters to improve the model's ability to reproduce local traffic conditions (Holm et al., 2007), whereas validation evaluates the calibrated model's capacity to accurately represent travel behavior using metrics independent from the calibration process, thereby supporting generalization and predictive capability.

**Table 9** summarizes the main parameters associated with each traffic model, although these are not exclusive to a particular model or software. Additional parameters include vehicle attributes and legal

**Table 9.** Metrics used for calibrating traffic models.

| Traffic model                        | Software                                                                                       | Parameter                                                                                                                                                         | Description                                                                                                                                                                                                                             |
|--------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gap acceptance                       | Based on Gipps (AIMSUM)                                                                        | Look ahead distance or Safety gap or safety distance or Headway or min distance between vehicles                                                                  | Necessary distance to avoid a collision if the leader decelerates heavily                                                                                                                                                               |
|                                      |                                                                                                | (Driver's) Reaction time or Time headway                                                                                                                          | Can be the same for all drivers (macro) or specific per type of vehicle, per level of congestion (micro), for example. It influences density and flow.                                                                                  |
|                                      |                                                                                                | (Driver's) magnitude of the reaction (acceleration, deceleration, or retardation rates)                                                                           | Can influence travel time delay and average speed                                                                                                                                                                                       |
|                                      |                                                                                                | Car-following parameters ( $\alpha$ , $\beta$ , $\gamma$ )                                                                                                        | Proportionality ratios between acceleration and speed, speed difference between follower and leader, and space headway.                                                                                                                 |
|                                      |                                                                                                | Max waiting time, Standstill distance, Speed model, critical gap, follow up gap.                                                                                  | The gaps can influence the maximum capacity of a subordinate flow within a node. The max waiting time is the longest time a vehicle of subordinate flow can wait to enter a node.                                                       |
| Psycho-physiological driver behavior | Based on GHR models (MITSIM) SUMO VISSIM Based on Fritzsche (PARAMICS) Based on Wiedeman model | Action points                                                                                                                                                     | Thresholds where the driver changes his/her behavior.                                                                                                                                                                                   |
|                                      |                                                                                                | Threshold for perception of negative and positive speed differences                                                                                               | Below these thresholds, the follower doesn't perceive the speed differences.                                                                                                                                                            |
|                                      |                                                                                                | Desired time gap, risky time gap                                                                                                                                  | Required to compute the thresholds: Desired distance, risky distance, safe distance, and braking distance.                                                                                                                              |
|                                      |                                                                                                | Safety time gap                                                                                                                                                   | Required to compute the thresholds: Desired distance between stationary vehicles, desired min following distance at low-speed differences, max following distance, approaching point, and decreasing, and increasing speed differences. |
|                                      |                                                                                                | Several fix and random numbers for different thresholds.                                                                                                          | Parameters required to simulate one-lane highway traffic model with the Cellular automaton model.                                                                                                                                       |
| Cell based model                     | MATSIM                                                                                         | Length of the road, length of the cell, probability of decrease speed, max speed, probability of switch lane, initial density, number of cars, acceleration rate. |                                                                                                                                                                                                                                         |
| Trajectory based model               |                                                                                                | Time lag, Distance lag<br>Desired speed, Desired time gap, Minimum gap, Max acceleration, Comfortable deceleration                                                | Required by the IDM model.                                                                                                                                                                                                              |

(Olstam and Tapani, 2004; Sun et al., 2020)

or desired driving limits. **Olstam and Tapani (2004)** provide an extensive description of the parameters commonly used in the first two traffic model categories presented in **Table 9**.

### 3.5.1. Goodness-of-fit Measures

Although earlier studies often limited validation to visual inspection of metric behavior without quantitative assessment of differences (e.g., **Ding, 2011; Olstam and Tapani, 2004**), current computational capabilities enable the systematic evaluation of multiple error functions.

During validation, model parameters may undergo optimization procedures (**Treiber and Kesting, 2013**), based on design of experiments, trial-and-error calibration, or sensitivity analysis (**Azam et al., 2019; Kanagaraj et al., 2013**). These approaches seek goodness-of-fit measures that minimize error functions, satisfy threshold criteria, or support statistical hypothesis testing. Error is defined as the difference between simulated outputs and observed data. Commonly used measures include the GEH statistic for traffic volumes; absolute and relative errors (**Giraldo et al., 2025; Hale et al., 2015; Krajewicz et al., 2002; Ranjitkar et al., 2005**), cosine similarity (**Giraldo et al., 2025**), mean absolute percentage error (MAPE), root mean square error (RMSE), (**Patwary et al., 2021; Treiber and Kesting, 2013**), and coefficient of determination ( $R^2$ ). **Table 10** describes the most common and recommended error functions in traffic simulation, where  $y^o$  is the observed value and  $y^s$  the simulated value.

The literature highlights several key considerations for calibration processes: (a) prioritizing global rather than purely local calibration based on gap measures; (b) selecting intuitive and plausible parameter values, preferably initialized from published data; (c) using data from representative locations rather than inflection or boundary points; and (d) beginning with zero-conflict conditions or scenarios without flow on the major road (**Sun et al., 2020; Treiber and Kesting, 2013**).

### 3.5.2. Model Performance Metrics

The suitability of a simulation model depends on its intended purpose, which determines the performance metrics to be monitored, calibrated, and estimated throughout the simulation process. Without aiming to be exhaustive, **Table 11** summarizes the most common output metrics, together with their level of detail and application scale. Among these, travel time, travel speed, density, and saturation stand out due to their broad applicability across

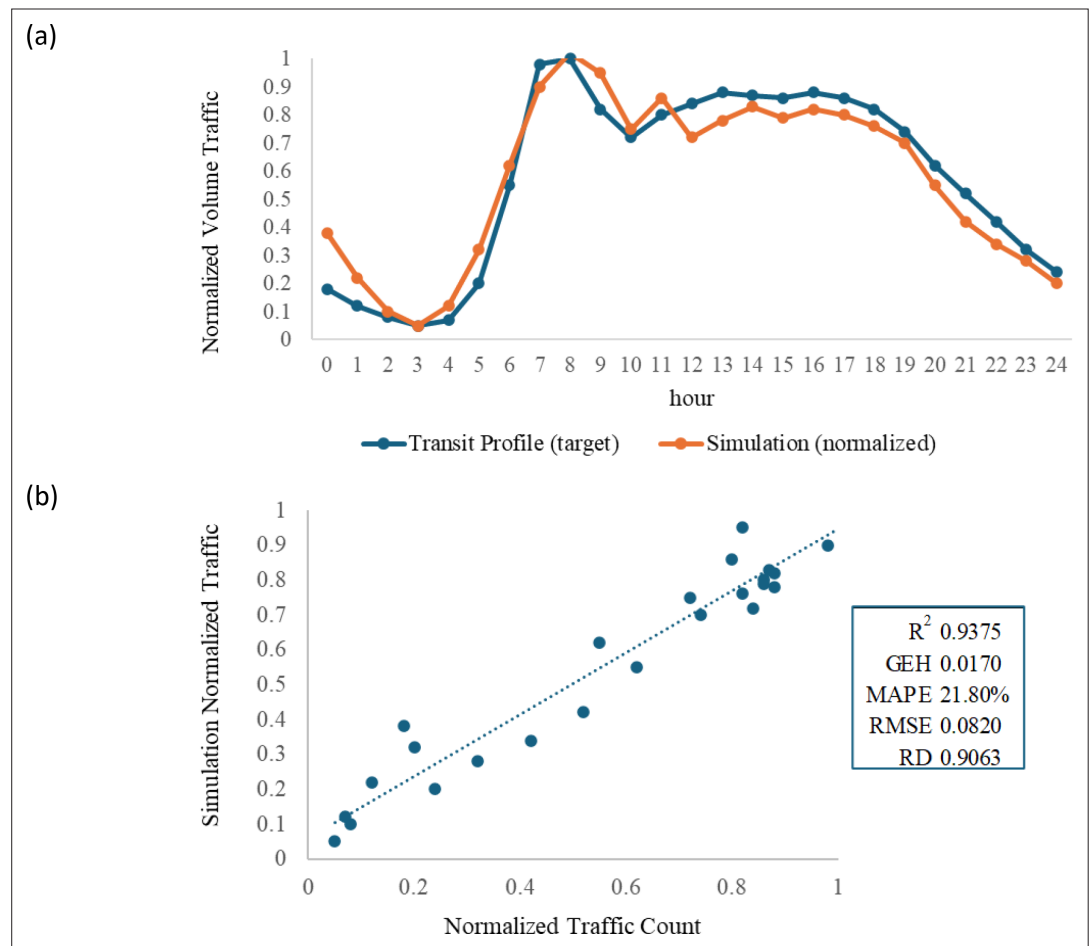
**Table 10.** Recommended error functions in traffic simulation

| Function                                    | Description                                                                              | Formula                                                       | Recommendation / Assumption/Strength                                                                    |
|---------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| $R^2$<br>Coefficient of determination       | Indicates how well the simulated variable explains the real one.                         | $R^2 = 1 - \frac{\sum (y^o - y^s)^2}{\sum (y^o - \bar{y})^2}$ | Assumes linear correlation.<br>Easy to understand, and common [0, 1]                                    |
| GEH statistic<br>(Geoffrey E. Havers)       | Normalizes the relative differences with respect to the magnitude of the compared flows. | $GEH = \sqrt{\frac{2 \sum (y^o - y^s)^2}{y^o + y^s}}$         | Mitigates the misleading effects that relative differences may produce.<br><5: good fit, > 10: poor fit |
| MAPE<br>Mean absolute percentage error      | Indicates the average percentage difference between observed and predicted values.       | $MAPE = \frac{100}{n} \sum_{i=1}^n \frac{ y^o - y^s }{y^s}$   | Easy to interpret as a percentage<br>[0 – 100]<br><10%: good fit                                        |
| RMSE<br>Root Mean Square Error              | Statistical measure that estimates the standard deviation of the errors distribution.    | $RMSE = \sqrt{\frac{\sum (y^o - y^s)^2}{n}}$                  | In the same units as the variable. Not a standard range.<br>Penalizes large errors.                     |
| Relative differences with Cosine similarity | Similarity between two vectors or tensors by means of the dot product between them.      | $RD = 1 - \frac{ y^o, y^s }{\ y^o\  \ y^s\ }$                 | For multi-dimensional variables.<br>Easy to read. Range [0, 1]<br>>0.9: good fit                        |

**Table 11.** Model performance metrics

| Metric                            | Units                                                      | Description                                                                                                                        | Detail |    |    | Scale |   | Purpose |   |
|-----------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|----|----|-------|---|---------|---|
|                                   |                                                            |                                                                                                                                    | Mi     | Me | Ma | N     | I | I       | O |
| Travel time                       | min/veh sec/<br>veh                                        | Average travel time of all vehicles of the simulation or in a specific facility or trajectory.                                     | x      | x  | x  | x     |   | x       | x |
| Travel Speed                      | km/h                                                       | Rate of motion (expressed in distance per unit of time)                                                                            | x      | x  | x  | x     |   | x       | x |
| Travel Distance                   | km/veh                                                     | Average extent of the space between the trip origin and the destination, measured along a vehicular route.                         | x      | x  | x  | x     |   | x       | x |
| Delay                             | Sec or min                                                 | Additional travel time experienced by travelers at speeds less than the free flow (posted) speed.                                  | x      | x  |    | x     |   |         | x |
| Speed ratio                       | -                                                          | Degree of traffic flow through intersections (speed adjacent to the access point over speed along the corresponding road section). | x      | x  | x  | x     |   |         | x |
| Stopping frequency                | % Stops per time-space bin                                 | Average relative frequency of stops per time, space, or segment bin. Shown in histograms.                                          | x      | x  |    | x     |   |         | x |
| Volume of Traffic or Traffic Flow | % Vehicles or persons per time-space bin                   | Number of persons or vehicles passing a point (or intersection) on a roadway per time interval. Shown in histograms.               |        |    | x  | x     |   | x       | x |
| Congestion of streets             | sec or min                                                 | Time lost per trips, level of congestion per road.                                                                                 |        |    | x  | x     |   | x       | x |
| Saturation (flow) rate            | % Vehicles per time bin                                    | Maximum rate of flow of traffic in a road segment per time interval.                                                               | x      | x  | x  | x     |   | x       | x |
| Density                           | veh/km or veh/<br>km-lane or<br>% Capacity of<br>lane/road | Number of vehicles on the roadway segment averaged over space.                                                                     | x      | x  | x  | x     |   | x       | x |
| Queue Length                      | km or<br># Vehicles                                        | Length of queued vehicles waiting to be served by the system.                                                                      | x      | x  |    | x     |   |         | x |
| Mode Split                        | % Persons                                                  | Percentage of travelers using each travel mode (SOV, HOV, transit, bicycle, pedestrian, etc.).                                     |        |    |    | x     |   | x       | x |
| Time-Space Diagrams               | km/h per (time-space bin)                                  | 3D graphs or surface response plots showing the speed at various times and spaces (positions).                                     |        |    |    | x     |   |         | x |
| Time-Flow Diagrams                | % of maximum flow                                          | Graphs show the flow behavior at various times with different densities.                                                           |        |    |    | x     |   |         | x |
| Cost of travel (due to delays)    | Time                                                       | The sum of the volume delay, turn penalty and junction delay functions.                                                            | x      |    | x  | x     |   |         | x |
| Cost of travel                    | \$ or time                                                 | Dynamic cost functions depend on the experienced travel time.                                                                      | x      | x  |    | x     | x |         | x |

Mi: Microscopic or specific, Me: Mesoscopic, Ma: Macroscopic or network-wide, I: Input, O: Output, N: Network-wide, I: vehicle or user specific.



**Figure 5** . Validation example based on traffic volume profiles using five goodness-of-fit measures.

different purposes and scales, as well as their availability in most simulation software. The units are given in SI units. Some of the metrics described in **Table 11** may also serve as input data for the calibration process.

As an example of the application of these metrics in a validation exercise, **Figure 5** presents the normalized traffic volume profile obtained from vehicle counts and from the corresponding traffic simulation model. **Figure 5a** illustrates the temporal profiles, while **Figure 5b** presents the correlation diagram between both variables, together with the metrics summarized in **Table 10**.

### 3.6. Analysis of Scenarios

Simulation scenarios represent variations of the current state of the modeled system, typically defined from a validated baseline scenario, and are used to assess implementation feasibility, roadway resilience, or the effects of population growth. Simulation outputs may be analyzed at the network level, aggregated by roadway or zone, or at the level of individual vehicles. These outputs quantify changes in system performance resulting from the evaluated scenarios and should remain comparable to the validation data collected during the initial stages of the methodology. Examples of simulation outputs are summarized in **Table 12**.

## 4. Implementation Recommendations

Experience with modeling and simulation using SUMO has highlighted several factors that can substantially affect performance and execution. Efforts to improve realism and reproduce the intrinsic characteristics of mobility patterns generally require modifications in three main areas: network configuration, additional coding for trip generation, and increased framework complexity.

**Table 12.** Examples of expected results from different scenarios

| Simulated scenario element modified | Examples                                                                                                                                                                                   | Expected results                                                                                               | Examples of Sources                             |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Network                             | Add/remove stoplights, lanes, entrances, intersection designs, etc.<br>Change of lane uses (bikes per vehicles)<br>Different modal orientations of integrated urban transportation systems | Changes in congestion, in travel times, in transferring time, integration effectiveness.                       | (Fierek and Zak, 2012)                          |
| Vehicle                             | Change buses for electric buses<br>Penetration of connected and automated vehicles                                                                                                         | Changes in emissions<br>Changes in congestion and network capacity<br>Changes in accident rate and road safety | (Raju and Farah, 2021)                          |
| Traffic flow management             | Lane sorting strategies<br>Converting HOVs into CACC (Cooperative Adaptive Cruise Control) lanes<br>Cooperative ITS (Intelligent Transportation Systems)                                   | Changes in traffic streams, operational traffic performance                                                    | (Codeca and Härrri, 2018; Raju and Farah, 2021) |
| Policy implementation               | Floating cars versus local taxis fleets                                                                                                                                                    | Differences in passenger wait time, pickup trip time                                                           | (Maciejewski et al., 2016)                      |

## 4.1. Network improvements

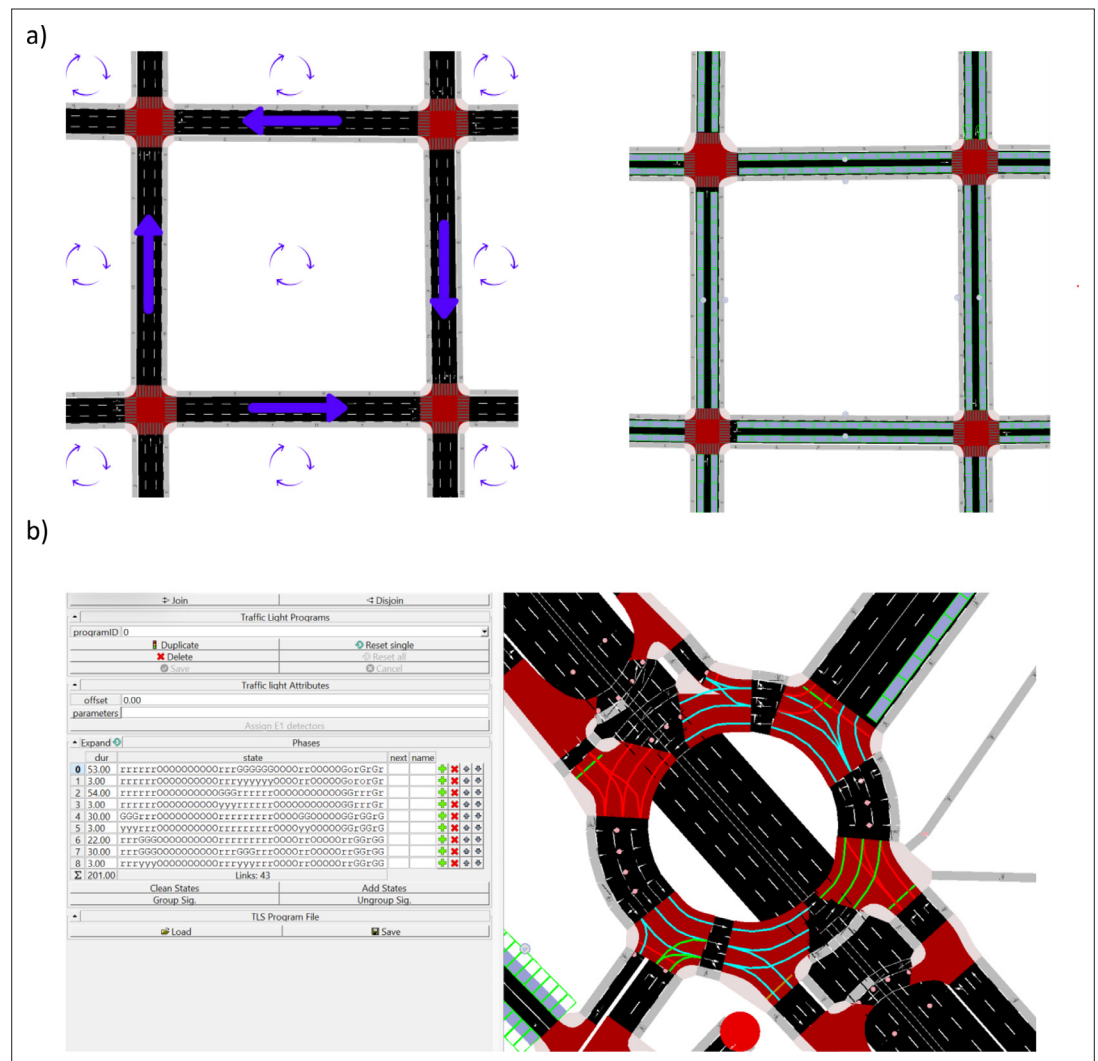
The network model should accurately represent the mobility patterns of the study area and capture key characteristics such as speed, congestion levels, and primary routing structures. Although simulations can still be executed without an ideal model, the resulting outputs may become difficult to validate against real-world data. During initial runs, execution failures may occur, and the presence of errors or warnings can substantially increase computation time, often resulting in incomplete or unsuccessful simulations.

**Pedestrian Path Flow:** When importing maps from GPS-based sources such as OpenStreetMap, pedestrian routes are often assigned to edges as unidirectional elements. On one-way roads, this causes both sidewalks to inherit the same pedestrian direction, potentially generating errors, warnings, and pedestrian congestion during simulation. To avoid these issues, all edges should include bidirectional pedestrian paths and maintain proper connectivity with adjacent edges to ensure continuous routes. **Figure 6a** illustrates this adjustment. Without these modifications, simulations may experience significant pedestrian jams and execution errors.

**Parking Stations Definition:** Parking stations should be defined as having realistic capacities and accurate locations across both commercial and residential areas. Although simplifying them as indefinite points may be acceptable for targeted simulations, it can introduce bias and inconsistencies in broader analyses. While a detailed definition increases modeling complexity and computation time, it provides higher-resolution and more reliable results.

**Traffic Light Programs:** Traffic signal timing and prioritization strongly influence simulation outcomes. Modifications to traffic light programs can increase traffic density, generate congestion, and extend computation time. Accurate traffic signal data obtained from official sources or field measurements are therefore recommended to ensure reliable simulation performance. **Figure 6b** presents an example of a detailed signal-controlled roundabout intersection.

**Intersection and Lane Maneuvers:** Lanes at intersections should be reviewed and adjusted to reflect local traffic conditions and right-of-way priorities. Imported network maps frequently lack detailed representations, often simplifying intersections into basic configurations. In addition, parking station locations may impose restrictions on adjacent lanes, which should be considered to ensure realistic simulation behavior.



**Figure 6 .** Examples of improvement implementations in the network.

(a) Pedestrian path flows, (b) Controlled intersections.

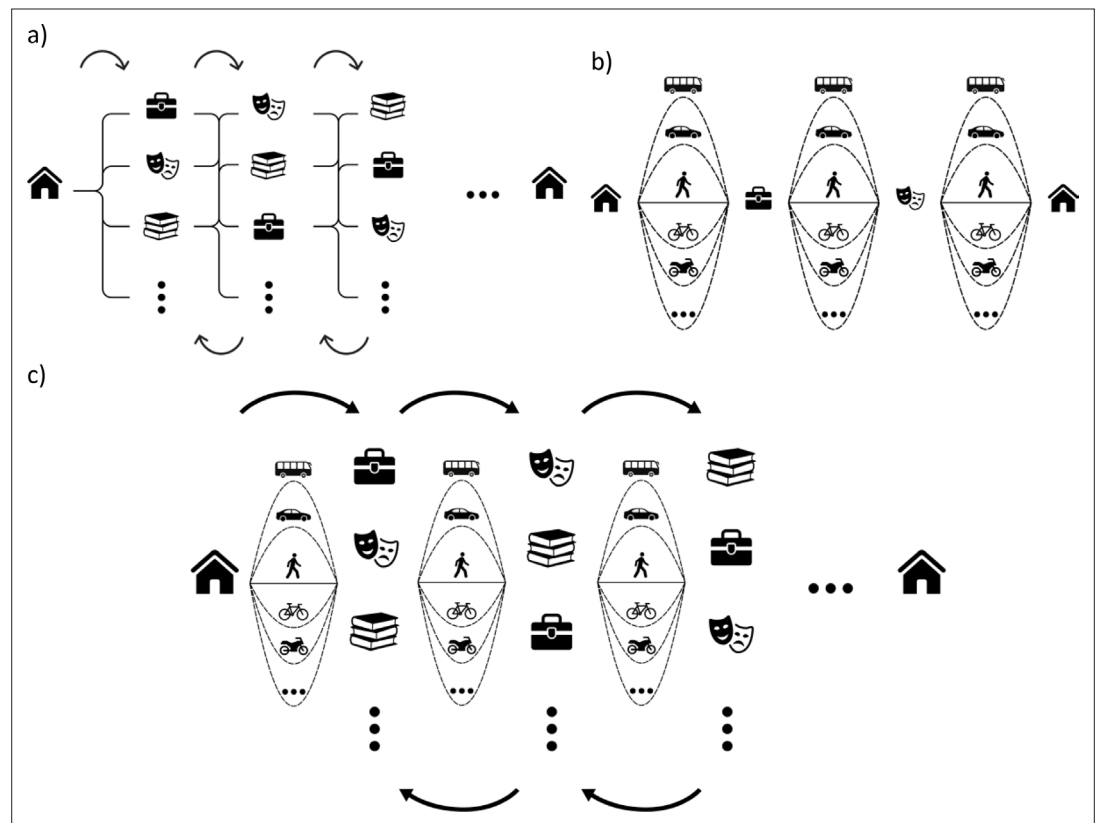
## 4.2. Activity-based trips implementation

Activity-based trip generation is widely used for modeling traffic demand by generating trips for individual mobility actors according to activity distributions and modal shares. Although conceptually straightforward, the complexity of activity chains depends on the desired level of resolution.

Complex activity chains remain less frequently addressed in the literature. In SUMO, the SAGA model presents limitations in scripting and trip-file generation, complicating the implementation of chains that combine multiple activities, stops, or transport modes. These chains may require returning to previous locations or ending at specific destinations. Nevertheless, they can represent realistic daily routines; for example, an adult may drop off children at school, commute to work, leave during the day to pick up the children, and then return to work. The required level of detail depends on the intended modeling precision. **Figure 7a** presents examples of these activity chains.

Incorporating multiple transport modes within an activity chain further increases variability across trip segments, as illustrated in **Figure 7b**. Although individual short trips may rely on different modes of transport, the overall modal distribution should remain consistent throughout the simulation.

Implementation challenges remain significant when defining parameters for activity chains that combine multiple activities and transport modes, as illustrated in **Figure 7c**. Even with advanced AI-assisted programming and coding tools, configuring these interactions continues to be a time-consuming and complex task.



**Figure 7** . Schemes of activity-based trips.

(a) activity chains considering work, study and entertainment. (b) multimodal activity chains. (c) combined multimodal and multi-activity chains.

### 4.3 Simulation Size

The spatial extent of a simulation influences the choice of modeling approach and software. While larger, detailed areas provide a better understanding of complex traffic phenomena, computational constraints and available resources limit the feasible size and scope.

**Micro-Scale Simulation:** Micro-scale simulations generally focus on limited areas, such as individual streets or neighborhoods, involving relatively few intersections, trips, and agents. As network complexity increases, accurate information on traffic signal programs, lane configurations, and intersection maneuvers becomes increasingly important. In SUMO, the NetEdit module supports the selection, modification, and definition of these elements.

**Large-Area Simulation Challenges:** Generating detailed trips over a large area can be computationally intensive or infeasible. The SAGA framework supports pre-generation of trip plans, allowing the simulation to focus on trip execution.

**Parallel Processing:** Recent SUMO developments incorporate multiprocessing capabilities for operations such as parking station rerouting, substantially reducing computation time in networks with extensive parking infrastructure.

**Boundary Effects:** Restricting the analysis to a sub-region may introduce inconsistencies associated with incomplete trips crossing the simulation boundary. Trips can be classified as: (1) entirely within the study area, (2) originating within the area and leaving it, (3) entering the area and remaining within it, or (4) traversing the area. Proper classification is essential to preserve realism in micro-scale simulations.

Overall, these techniques and recommendations support a more streamlined simulation workflow and facilitate the extraction of meaningful insights from digital mobility models. Although specific software platforms are commonly associated with particular simulation scales, many tools can be

adapted across scales. Advances in computational capabilities and modeling methodologies increasingly enable the integration of multiple approaches to overcome individual limitations.

## 5. Conclusions

Traffic models are essential tools for representing and understanding vehicular movement in urban networks. They support the evaluation of mobility strategies, infrastructure optimization, and the assessment of environmental impacts such as energy consumption and emissions. This review consolidated the methodological and computational foundations of traffic microsimulation, synthesizing the key stages required for realistic mobility representation, from data collection to calibration and validation.

The proposed six-step framework provides a structured methodology adaptable to microscopic, mesoscopic, and macroscopic scales according to data availability, computational resources, and decision-making objectives. It integrates multimodal and activity-based approaches to improve the representation of real-world mobility dynamics. The comparative assessment of simulation platforms highlights complementary strengths among existing tools. Open-source platforms such as SUMO and MATSim provide flexibility and scalability, whereas commercial tools including PTV VISSIM, PTV VISUM, and Aimsun offer advanced visualization and calibration capabilities. Recent integrations with platforms such as Unity, CARLA, and Python APIs continue expanding the analytical and visualization capabilities of urban mobility models.

Overall, microsimulation has evolved beyond the replication of traffic flows into a robust framework for sustainable transportation planning and policy evaluation. Advances in digital twins, real-time data assimilation, and AI-driven optimization are further strengthening its role in the development of resilient and intelligent transportation systems.

Future research should address several emerging challenges, including advanced data-driven methods for configuring complex activity chains, the integration of individualized data from personal sensing devices to improve dynamic travel demand estimation, and the use of parallel and next-generation computing approaches to support larger and more detailed simulations. Incorporating daily variations in traveler behavior and demand will also be essential for improving the predictive and prescriptive capabilities of future microsimulation frameworks.

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## Acknowledgements

Special thanks to RELIEVE, the Latin American Network of Researchers in Energy and Vehicles, for supporting the collaboration among the research team.

## Funding

No external funding for this article is reported. Internal Grant: This work was partially funded by the Challenge-Based Research Funding Program (Tecnologico de Monterrey). Grant 5607TM-10-393 MAITEC.

## Conflict of Interest

No competing interests reported.

## Data and code availability

N/A

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## Appendix A

### Applications per Software

#### SUMO applications

Recent applications of SUMO in traffic research highlight its versatility. One study demonstrated how Unmanned Aerial Vehicles (UAV) data, when combined with SUMO, can capture the dynamics of traffic shockwaves at intersections. It revealed that rear-end collision risk propagates with these waves and concentrates at the rear of queues (**Wang et al., 2025**). Another investigation enhanced the realism of cyclist behavior by incorporating empirical insights from the SimRa dataset to design distinct cyclist profiles and improve intersection logic, thus overcoming the limitations of the default bicycle model (**Karakaya et al., 2023**). Large-scale applications are also evident. A co-simulation environment linking CARLA, an open-source simulator for autonomous driving research, with SUMO has been developed to evaluate the performance of automated vehicles under calibrated traffic demand, enabling robust system-level testing of safety and efficiency (**Yilmaz-Niewerth et al., 2024**). At the urban scale, commuting patterns in Aichi Prefecture were simulated using calibrated trip and vehicle data, successfully reproducing accident-related patterns and informing prevention strategies (**Suzuki et al., 2024**).

The combination of SUMO with external visualization engines has opened new ground in transport modeling. A framework integrating Unity 3D and Blender with OpenStreetMap data enables real-time visualization and multi-scale emission measurement (**Nagy et al., 2025**). On the methodological side, comparative analyses of demand generation tools embedded in SUMO have shown that tool selection directly influences network connectivity, emissions, and re-routing behavior, factors critical for reliable vehicular network simulations (**Barbecho Bautista et al., 2022**). Policy-oriented research has also expanded, demonstrating that congestion in Zurich can only be avoided if more than 50 % of car users shift to bicycles (**Fulton et al., 2025**). Similarly, analyses of urban form have revealed that rectangular block designs consistently outperform triangular and radial configurations in both efficiency and sustainability (**Essamlali et al., 2025**).

The integration of SUMO with external visualization engines and methodological tools has broadened its application in traffic research. Frameworks such as SUMITY enable real-time visualization and multi-scale emission assessment, while the choice of demand generation tools has been shown to affect network connectivity, emissions, and rerouting behavior. SUMO is increasingly applied in policy and planning studies, including evaluations of bicycle infrastructure and urban block design, demonstrating its value for assessing efficiency and sustainability across diverse urban contexts.

Energy and environmental dimensions have also been modeled in detail. The incorporation of fine-grained energy variables into SUMO-based bus simulations has demonstrated how stop spacing, vehicle type, and passenger load influence energy consumption, highlighting the framework's potential for digital-twin development (**García-Cerrud et al., 2024**). Integration of live video data with SUMO experiments has enabled quantification of heterogeneous transport behavior with over 95 % classification accuracy, demonstrating the benefits of linking field data and simulation for enhanced safety management (**Ravindran et al., 2025**). SUMO has also been connected with analytical hierarchy process methods to evaluate public acceptance of transport technologies, showing that NGVs lead in acceptance due to cost and infrastructure reliability. At the same time, EVs perform best environmentally but remain limited by affordability and network constraints (**España et al., 2025**).

Collectively, these studies illustrate the versatility of SUMO when combined with empirical data, co-simulation platforms, or decision-analysis tools. The open-source framework not only enhances methodological rigor but also supports practical insights for safer, greener, and more socially acceptable mobility systems.

## VISUM applications

Beyond open-source environments, proprietary tools like VISUM have also been extensively used to explore sustainable transport strategies, multimodal planning, and policy interventions. VISUM-based studies frequently integrate GIS data, participatory methods, and external models to support comprehensive decision-making. The mapping of transport externalities, such as emissions, noise, and congestion, has enabled the identification of critical blackspots for mitigation strategies (*Sampaio et al., 2022*). Bicycle traffic modeling in Warsaw has been incorporated into a broader regional transport framework to support cycling policy evaluation (*Jacyna et al., 2017*). Pedestrian-oriented analyses combining demand modeling, microscopic simulation, and participatory approaches have demonstrated the usefulness of multi-criteria decision processes for evaluating pedestrianization alternatives (*Gülhan et al., 2025*). Evaluations of Madrid's "Centro" low-emission zone have shown notable improvements in pedestrian air quality following car-access restrictions (*Sánchez et al., 2021*).

Further research has explored the link between emissions and urban form. Bottom-up emission modeling integrating the Irish National Transport Model with VISUM found that electric vehicle adoption yields greater CO<sub>2</sub> savings exits (*Charly and Caulfield, 2025*). Simulation of alternative morphologies in Aveiro, Portugal, indicated that compact urban layouts reduce emissions, whereas dispersed forms exacerbate traffic pollution (*Augusto et al., 2024*). Analyses of depopulation in Łódź, Poland, revealed that urban shrinkage promotes decentralization, longer commutes, and a weaker service core, with significant implications for planning (*Sahebgharani et al., 2024*).

Data-driven and digital integration approaches are increasingly prominent. The use of smartphone records for demand estimation has demonstrated the potential of mobile data to complement traditional surveys in scenario testing (*Montero et al., 2022*). Linking OD-matrix calibration and Python routines with VISUM has advanced digital-twin development and mobility replication for long-term decision-making (*Pala et al., 2025*). Comparative evaluations of static and dynamic tolling have shown that dynamic link-based schemes enhance welfare under autonomous and shared-autonomous vehicles, while static tolls perform better in mixed fleets (*Shatanawi et al., 2022*). Finally, multimodal scenario assessments along the Lao Cai–Hanoi–Hai Phong–Quang Ninh corridor have indicated that integrated strategies improve connectivity while reducing emissions (*Nguyen et al., 2025*).

Together, VISUM-based studies indicate that simulation is increasingly applied not only for technical modeling but also for evaluating policy instruments, sustainable designs, and social trade-offs. The findings show that VISUM, often integrated with complementary data sources and analytical methods, functions as a versatile decision-support tool across contexts ranging from emissions reduction and cycling promotion to pricing, pedestrianization, and corridor-level investment planning.

## VISSIM applications

Compared with open-source platforms, VISSIM is typically employed when greater microsimulation detail or explicit driver–vehicle interaction modeling is required. Applications span logistics, urban networks, safety, and behavioral analysis. City logistics modeling has demonstrated that intersection redesign can reduce congestion and promote sustainable freight operations (*Kučera and Chocholáč, 2021*). During the COVID-19 pandemic, simulations of Calgary's Marlborough transit station tested distancing rules and spatial redesigns, revealing that simple, low-cost interventions could balance infection risk and operational efficiency (*Miao and Saidi, 2025*). In the heterogeneous traffic conditions of Mangalore City, calibrated microsimulation identified short- and long-term measures that improved flow and reduced travel times (*Bandi and George, 2020*). They also evaluated the impact of vehicle and driver characteristics on the traffic volume prediction in the same city (*Bandi and George, 2020*). At the freeway level, the evaluation of ramp metering strategies in Thessaloniki confirmed that both fixed and actuated control can improve performance and reduce environmental impacts (*Mitkas and Politis, 2020*).

Recent developments include the "Virtual Emergency Lane" (VEL) strategy, which reallocates lanes dynamically during incidents to enable rescue vehicle access, yielding

reductions in delay and improved emergency response efficiency (**Wang et al., 2025**). Broader methodological reviews have summarized advances in driving-behavior simulation using artificial intelligence, covering calibration techniques, car-following models, heterogeneous traffic, and AI-based optimization of driver responses (**Al-Msari et al., 2024**). Pedestrian–vehicle conflict modeling has also benefited from VISSIM’s integration with the SSAM safety assessment tool, achieving over 90% prediction accuracy and demonstrating that mitigation measures can reduce conflicts by up to 51% (**Hussain et al., 2024**). Collectively, these studies demonstrate that VISSIM has evolved into a flexible platform for analyzing city logistics, public transit resilience, roadway performance, incident response, and safety evaluation—particularly when combined with AI or surrogate safety models.

Overall, the growing body of research using VISSIM underscores its strength in detailed, behavior-based simulation, where human decision-making, congestion dynamics, and safety risks are central. Unlike open-source frameworks, VISSIM allows fine calibration of driver, vehicle, and infrastructure interactions, making it a preferred tool when precision in microscopic modeling is required, especially in studies linking operational performance to safety and emergency management.

## MATSim applications

In contrast, MATSim focuses on multi-agent, activity-based simulation, offering a powerful environment for evaluating policies, demand-responsive transport, and the integration of emerging mobility services. Recent work has demonstrated MATSim’s evolution through co-simulation, extensions, and large-scale urban applications. Integrated frameworks combining MATSim with external fleet simulators have enabled realistic modeling of ride-pooling and heterogeneous services without additional extensions (**Yang et al., 2024**). A validated methodology for cycling in England incorporated a “quietness” attribute to capture environmental effects on route choice, providing results transferable to other regions (**Alvarez Castro et al., 2024**). Enhancements to the MATSim Open Berlin scenario improved activity-based demand generation, automated calibration, and included commercial traffic, increasing model realism relative to earlier versions (**Rakow et al., 2025**). Automated generation of lanes and traffic signals from OpenStreetMap has further streamlined modeling and enabled the testing of adaptive signal strategies (**Ziemke and Braun, 2021**).

Additional studies have employed MATSim to assess on-demand mobility solutions. Prebooking mechanisms have been introduced into demand-responsive transport simulations (**Hörl et al., 2023**), and reinforcement learning approaches such as Q-learning have been applied to fleet rebalancing (**Chouaki et al., 2022**). Further developments include modeling peer-to-peer car sharing through synchronization concepts for limited vehicle and parking resources, tested in a Berlin case study (**Hörl et al., 2024**). Integration of MATSim with the FEATHERS activity-based model has expanded its policy scope to encompass activity participation and location choice (**Ziemke et al., 2021**).

Taken together, these studies demonstrate that MATSim functions as a highly adaptable simulation ecosystem extending far beyond traditional traffic flow analysis. From cycling and pedestrian policy assessment to digital-twin development and fleet-based mobility services, MATSim enables detailed representation of behavioral dynamics, incorporation of advanced algorithms, and evaluation of long-term sustainability strategies. Its agent-based foundation makes it a key tool for linking individual decision processes with large-scale mobility planning.